

**AD23431 - STATISTICAL ANALYSIS AND COMPUTING****EXP 2: Implementation of Vectors, Matrices, and Arrays in R****I Electricity Consumption Comparison Using Matrices**

A college records electricity consumption (in units) for 4 hostels during Morning, Afternoon, and Night.

Using Matrices in R, do the following:

- a. Create a matrix where rows represent hostels and columns represent time slots.

```
consumption <- matrix(  
  c(121, 152, 103,  
    134, 165, 116,  
    147, 178, 129,  
    121, 152, 103),  
  nrow = 4,  
  byrow = TRUE  
)  
rownames(consumption) <- c("Hostel A", "Hostel B", "Hostel C",  
  "Hostel D")  
colnames(consumption) <- c("Morning", "Afternoon", "Night")  
consumption
```

Output:

```
> consumption  
Morning Afternoon Night  
Hostel A    121      152   103  
Hostel B    134      165   116  
Hostel C    147      178   129  
Hostel D    121      152   103
```

- b. Compute the total electricity consumption per hostel.

```
total_per_hostel <- rowSums(consumption)  
total_per_hostel
```

Output:

```
Hostel A Hostel B Hostel C Hostel D  
376     415     454     376
```



- c. Identify the time slot with the highest overall consumption.

```
total_per_time <- colSums(consumption)
```

```
total_per_time
```

```
highest_time_slot <- names(which.max(total_per_time))
```

```
highest_time_slot
```

Output:

```
> total_per_time
```

```
Morning Afternoon Night
```

```
523 647 451
```

```
> highest_time_slot
```

```
[1] "Afternoon"
```

- d. Normalize the matrix values by dividing each entry by the maximum consumption value.

```
max_value <- max(consumption)
```

```
normalized_consumption <- consumption / max_value
```

```
normalized_consumption
```

Output:

```
> normalized_consumption
```

```
Morning Afternoon Night
```

```
Hostel A 0.6797753 0.8539326 0.5786517
```

```
Hostel B 0.7528090 0.9269663 0.6516854
```

```
Hostel C 0.8258427 1.0000000 0.7247191
```

```
Hostel D 0.6797753 0.8539326 0.5786517
```

- e. Convert the matrix into a data frame with appropriate row and column names.

```
consumption_df <- as.data.frame(consumption)
```

```
consumption_df
```

Output:

```
> consumption_df
```

```
Morning Afternoon Night
```

```
Hostel A 121 152 103
```

```
Hostel B 134 165 116
```

```
Hostel C 147 178 129
```

```
Hostel D 121 152 103
```



II Manufacturing Cost and Profit Analysis

A factory produces three products across four production units.

Two matrices are maintained:

a. Cost Matrix (production cost)

b. Revenue Matrix (selling revenue)

c. Create cost and revenue matrices.

```
cost_matrix <- matrix(c(10, 12, 11, 15, 20, 22, 25, 21, 30, 35, 32, 33),  
nrow = 3, byrow = TRUE)
```

```
rev_matrix <- matrix(c(15, 18, 16, 20, 30, 32, 35, 31, 45, 50, 48, 47),  
nrow = 3, byrow = TRUE)
```

```
rownames(cost_matrix) <- rownames(rev_matrix) <- c("P1", "P2",  
"P3")
```

```
colnames(cost_matrix) <- colnames(rev_matrix) <- c("Unit1", "Unit2",  
"Unit3", "Unit4")
```

Using Matrices in R, do the following:

d. Compute the profit matrix using matrix subtraction.

```
profit_matrix <- rev_matrix - cost_matrix  
print(profit_matrix)
```

Output:

```
> print(profit_matrix)  
      Unit1 Unit2 Unit3 Unit4  
P1      5    6    5    5  
P2     10    10    10    10  
P3     15    15    16    14
```

e. Calculate total profit per product.

```
total_profit_product <- rowSums(profit_matrix)  
print(total_profit_product)
```

Output:

```
> print(total_profit_product)  
P1 P2 P3  
21 40 60
```



- f. Determine the production unit with maximum profit.

```
total_profit_unit <- colSums(profit_matrix)  
max_unit <- names(which.max(total_profit_unit))  
print(paste("The production unit with maximum profit is:", max_unit))
```

Output:

```
[1] "The production unit with maximum profit is: Unit2"
```

- g. Verify whether the cost and revenue matrices are conformable for subtraction.

```
is_conformable <- all(dim(cost_matrix) == dim(rev_matrix))  
if(is_conformable) {  
  print("Matrices are conformable for subtraction.")  
} else {  
  print("Matrices are NOT conformable.")  
}
```

Output:

```
[1] "Matrices are conformable for subtraction."
```



III Traffic Flow Analysis Using 3D Arrays

Traffic data is collected for 3 intersections, 7 days, and 24 hours.

Using Arrays in R, do the following:

- a. Create a 3D array to store traffic counts.

```
data_vals <- c(10, 20, 30, 15, 25, 35, 20, 30, 40, 50, 60, 70, 55, 65, 75, 60, 70, 80)
```

```
traffic_array <- array(data_vals, dim = c(3, 2, 3),  
                        dimnames = list(Intersection = c("Int_A", "Int_B",  
"Int_C"), Day = c("Sat", "Sun"), Hour = c("H1", "H2", "H3")))  
print(traffic_array)
```

Output:

```
, , Hour = H1
```

```
      Day  
Intersection Sat Sun  
Int_A  10  15  
Int_B  20  25  
Int_C  30  35
```

```
, , Hour = H2
```

```
      Day  
Intersection Sat Sun  
Int_A  20  50  
Int_B  30  60  
Int_C  40  70
```

```
, , Hour = H3
```

```
      Day  
Intersection Sat Sun  
Int_A  55  60  
Int_B  65  70  
Int_C  75  80
```

- b. Calculate the average traffic per intersection.

```
avg_traffic <- apply(traffic_array, 1, mean)  
avg_traffic
```

Output:

```
Int_A Int_B Int_C  
35    45    55
```



- c. Identify the day with peak traffic across all intersections.

```
daily_totals <- apply(traffic_array, 2, sum)  
daily_totals  
peak_day <- names(which.max(daily_totals))  
peak_day
```

Output:

```
> daily_totals  
Sat Sun  
345 465  
> peak_day  
[1] "Sun"
```

- d. Extract traffic data for a specific intersection on weekends.

```
int_a_weekend <- traffic_array[1, , ]  
print(int_a_weekend)
```

Output:

```
Hour  
Day H1 H2 H3  
Sat 10 20 55  
Sun 15 50 60
```

- e. Reduce the array to a 2D matrix showing daily totals per intersection.

```
daily_matrix <- apply(traffic_array, c(1, 2), sum)  
colnames(daily_matrix) <- c("Sat_Total", "Sun_Total")  
print(daily_matrix)
```

Output:

```
Day  
Intersection Sat_Total Sun_Total  
Int_A      85      125  
Int_B     115     155  
Int_C     145     185
```



IV Student Attendance Monitoring Using Arrays

Attendance is recorded for 5 students, 4 subjects, and 30 days.

Using Arrays in R, do the following:

- Create a 3D attendance array (Present = 1, Absent = 0).

```
attendance_data <- c(1,1,0, 1,0,1, 1,1,1, 1,1,0, 1,0,0, 1,0,0, 1,1,0, 0,1,0)
attn_array <- array(attendance_data, dim = c(3, 2, 4),
dimnames = list(Student = c("S1", "S2", "S3"),
Subject = c("Math", "Sci"),
Day = paste0("D", 1:4)))
```
- Compute attendance percentage per student.
- Identify students who have less than 75% attendance.
- Extract attendance for a single subject across all students and days.
- Convert daily attendance into monthly subject-wise summaries.